

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

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Finding Reasoning Paths :

1. **Gestational diabetes**: Maternal diabetes is a known risk factor for neonatal complications, including respiratory distress syndrome (RDS) due to surfactant deficiency.
2. **Premature birth**: The baby was born after the 32nd gestational week, which is considered late preterm (34 - 36 weeks), increasing the risk of RDS.
3. **Cyanosis**: Cyanosis with no significant improvement despite oxygen therapy suggests hypoxemia due to a respiratory or pulmonary cause.
4. **Tachypnea and retractions**: Increased respiratory rate and intercostal retractions are hallmark signs of respiratory distress.
5. **Heart rate**: An elevated heart rate (104/min) may reflect compensatory mechanisms for hypoxemia or acidemia.
6. **Respiratory distress syndrome**: This is the most common cause of respiratory distress in preterm infants, often due to surfactant deficiency.

Reasoning Process :

1. **Maternal diabetes**: Gestational diabetes can impair fetal lung development, leading to surfactant deficiency and an increased risk of RDS. This is a significant contributing factor in this case.
2. **Prematurity**: The baby was born at 32 weeks, which places them at high risk for RDS. Preterm infants often lack sufficient functional surfactant in their lungs, leading to alveolar collapse and respiratory distress.
3. **Cyanosis**: The persistent cyanosis despite oxygen therapy suggests significant hypoxemia. In preterm infants, RDS is the most common cause of hypoxemia due to impaired gas exchange.
4. **Tachypnea and retractions**: The elevated respiratory rate (tachypnea) and intercostal retractions indicate increased work of breathing, which are classic signs of respiratory distress in preterm infants.
5. **Heart rate**: The heart rate of 104/min is elevated and may be due to hypoxemia or compensatory mechanisms. This is not unusual in preterm infants with RDS.
6. **Differential diagnosis**: While other conditions such as transient tachypnea of the newborn (TTN) or congenital infections could cause respiratory distress, the combination of prematurity, maternal diabetes, and persistent cyanosis strongly points to RDS as the most likely diagnosis.
7. **Surfactant deficiency**: RDS in preterm infants is primarily caused by a deficiency in pulmonary surfactant. This deficiency leads to alveolar collapse, decreased lung compliance, and impaired gas exchange, which explains the cyanosis and tachypnea observed in this baby.

Conclusion :

The most likely diagnosis for this newborn is **neonatal respiratory distress syndrome (NRDS)**, which is closely associated with surfactant deficiency. This condition is a common complication in preterm infants, particularly those born at late preterm gestation.

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<answer>

Neonatal respiratory distress syndrome (NRDS)

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